Extended conceptual architecture for workload-specific infrastructure planning

Inputs and evidence

User requirements
Workload and deployment type
Deadline or throughput
Latency, accuracy and limits

Candidate catalogue
Observed configurations
Versioned hardware metadata
Missing fields remain missing

External evidence
MLPerf result records
OEM specifications
Comparable prices when verified
Provenance and retrieval date

Workload-specific decision engines

Workload profile and evidence boundary
Validate inputs and select only matching empirical records

Training feasibility
Time-to-quality
Deadline and limits
S1 evidence

Inference feasibility
Throughput and latency
Accuracy and limits
S2 evidence

Hard-constraint filter
Test every requirement
Retain rejection reasons
Return infeasible if empty

Explainable ranking
Minimum resources first
Declared tie-break
No unsupported price claim

Bounded supporting analyses

Infrastructure sizing
Accelerators, nodes and rack fit

Power and energy
Measured value or labelled proxy

Cost analysis
Only with comparable evidence

Evidence and risk log
Assumptions, gaps and violations

Decision outputs

Feasible recommendation
Configuration and justification

Infeasible outcome
Binding constraints explained

Decision package
Configuration and checks
Provenance and assumptions
Limitations and report fields

Conceptual extension — not validated in this study

Existing-facility profile
Current assets and operating state

Gap and compatibility
Compare current and recommended state

Upgrade alternatives
Roadmap and replacement options